

PAIR AHMED JIKO

Lab In-Charge (Biochemist) | Scientific Officer | Molecular Biologist

Research Assistant | [Department of Chemistry \(CUET\)](#)

1534/B; Sayed Shah Road; Chawkbazar; Chattogram 4203; Bangladesh

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Clinical Biochemist and Analytical Chemist with interdisciplinary academic training in Biochemistry and Biotechnology, Chemistry, and Public Health, supported by broad experience in clinical biochemistry, molecular diagnostics, phytochemistry, and biomedical research. Experienced in in-vivo and in-vitro investigations, phytochemical isolation and characterization, molecular biology, and laboratory quality management, with additional experience in academic teaching, internship supervision, and professional training. Driven by a strong interest in molecular mechanisms of health and disease, particularly cancer biology, genetics, immunology, molecular biology, and bioactive compounds. Seeking advanced research training in molecular and biomedical sciences to strengthen specialized research expertise and contribute to rigorous, interdisciplinary, and evidence-based scientific investigations.

RESEARCH INTERESTS

Medicinal & Analytical Chemistry, Cancer Biology & Oncology, Molecular Biology, Clinical Chemistry, Drug Discovery, Public Health, Disease & Translational Biomedical Science.

EDUCATIONAL QUALIFICATIONS

- **2025 • Masters in Chemistry**
Department of Chemistry
Chittagong University of Engineering Technology (CUET)
Thesis: [Isolation, Characterization and in silico Studies of Bioactive Terpenoid and Flavonoid from the Seeds of *Mucuna pruriens*.](#)
Supervisor: [Prof. Ranajit Kumar Sutradhar; PhD](#)
- **2024 • Master of Public Health (MPH)**
Study Centre: Chittagong Medical College and Hospital (CMCH)
School of Science and Technology; Bangladesh Open University (BOU)
Thesis: Molecular Epidemiology and Genetic Diversity of the Circulating Dengue Serotypes in Chattogram: 2024 Outbreak
Supervisor: [Prof. Sayeed Mahmud; PhD, MPH, BCS, MPhil](#)
- **2021 • BSc in Biochemistry and Biotechnology**
Faculty of Life Science
University of Science and Technology Chittagong (USTC)
Project: Implant Project Tour at The Albion Laboratories Limited: 2020 (A Pharmaceutical Company)
Supervisor: [Prof. Nurul Absar; PhD](#)

PROFESSIONAL EXPERIENCES

- Nov'2020 – Present • **Lab-Incharge (Biochemist)**
Department of Clinical Biochemistry
Institute of Applied Health Sciences Hospital-USTC (ex-BBMH)
USTC, Foy's Lake, Khulshi, Chattogram 4202, Bangladesh.
- Sep'25 – Mar'26 • **Scientific Officer (Quality Assurance)**
Department of Laboratory Medicine
Epic Health Care (ISO 15189:2022 accredited Lab)

- Jan'25 – Feb'26 • **Lab-Incharge**
Feni Health Care (Approved GAMCA/WAFID Medical Centre)
Total Tower, Firozshah H/A, DT Road, Cornelhat, CTG-4000
- Sep'21 – Aug'22 • **Molecular Biologist**
Chevron Covid-19 RT-PCR & Antigen Testing Lab
Chevron Clinical Laboratory (Pte) Ltd.
993/2121; Agrabad Access Road; Haliashahar (Shantibag Branch), Chattogram.
- Sep'2015 – Dec'2020 • **Senior Laboratory Technologist**
Life Save Blood Bank & Transfusion Centre
1284, O.R. Nizam Road, Probortakmor, Pachlaish, Chattogram.

RESEARCH ARTICLE

1. Chowdhury, S. N., Majumder, S., Joy, M. I. S., Azad, A. K., Subaika, A. H., Hassan, K., Barua, A., Mysha, R. Y., Chowdhury, U. T., Chowdhury, R., Dey, B., Ara, J., Khan, A. A., **Jiko, P. A.**, Ahmed, A., (2026). Formulation, Evaluation, and Antimicrobial Investigation of a Polyherbal Soap against Skin Pathogens. *Pre-Print*. DOI: 10.21203/rs.3.rs-10948270/v1
2. Nipa, F. A., **Jiko, P. A. *** (Corresponding author), Majumder, S., Joy, M. I. S., Mohajan, S., Chowdhury, S., Fahim, T. H., Dutta, M., Khalil, E., Wahab, I. B., Islam, M., (2026). Antidiabetic Potential of *Ficus hispida* Fruit Extract in Alloxan-Induced Swiss Albino Diabetic Mice: In Vivo and Molecular Docking Study. *Pre-Print*. DOI: 10.21203/rs.3.rs-10568437/v1
3. Shil, P., Mohajan, S., Majumder, S., Alam, S., Haider, A., Chowdhury, N., **Jiko, P. A. *** (Corresponding author) (2026). Immune Response and Molecular Insights of Dengue Fever: A Systematic Review. *Pre-Print*. DOI: 10.13140/RG.2.2.22546.62405
4. Rakib, T. M., Hossen, F. B., Chowdhury, S., **Jiko, P. A.**, Mohajan, S., Alam, S., Khan, A. A., Majumder, S., Absar, N., Siddiki, A., (2026). Co-occurrence of Carbapenemase Genes, Disinfectant Resistance Determinants, and Class 1 Integrons in *Acinetobacter* spp. Isolated from Bloodstream Infections in Critical Care Units. *Pre-Print*. DOI: 10.64898/2026.04.27.721256
5. Chowdhury, S. N., Majumder, S., Joy, M. I. S., Azad, A. K., Subaika, A. H., Hassan, K., Islam, M. M., **Jiko, P. A. *** (Corresponding author) (2026). Hypomagnesemia and its clinical outcome: a prospective observational study in a tertiary care hospital in Bangladesh. *Annals of Medicine & Surgery*. DOI: 10.1097/MS9.00000000000005242
6. Chowdhury, K. H., Hasan, R., **Jiko, P. A.**, Karim, S., (2026). Comparative Study Analysis of Reheated Local Unrefined Oils and Store-Bought Refined Oil in Wister Albino Rats. *Journal of Applied Life Sciences International*. DOI: 10.9734/jalsi/2026/v29i3776
7. Tofazzal, R., Hossen, B., Chowdhury, S., **Jiko, P. A.**, Mohajan, S., Alam, S., Khan, A. A., Majumder, S., Arafat, M., Absar, N., Siddiki, Z (2026). Integron-Mediated Convergence of Carbapenemase and Disinfectant Resistance in 2 *Acinetobacter* spp. from Critical Care Units. *Pre-Print*. DOI: 10.64898/2026.04.27.721256
8. Sutradhar, R. K., **Jiko, P. A.**, Hossain, M. S., Islam, M.N., Hossain, M. E. (2026). Isolation, Characterization and in silico Studies of Bioactive Terpenoid and Flavonoid from the Seeds of *Mucuna pruriens*. *Asian Journal of Chemistry*. DOI: 10.14233/ajchem.2026.35289.
9. Sutradhar, R. K., Hossain, M. S., **Jiko, P. A.**, Islam, M.N., Hossain, M. E. (2025). Isolation, Characterization and Molecular Docking Studies of Bioactive Phyto-constituents from the Seeds of *Mucuna pruriens*. *Oriental Journal of Chemistry*. DOI: 10.13005/ojc/410528.

10. Mohammad, M., Richi, T., Hossain, R., Arafat, M., **Jiko, P. A.**, Emon, N., Rabbi, S., Khastagir, T., Hossain, H., Alam, S. (2025). A hepatoprotective experiment on taro vegetable (*Colocasia esculenta* (L.) Schott) flower employing animal models by mitigating oxidative stress. *Animal Models and Experimental Medicine*. DOI: 10.1002/ame2.70031.
11. Mohammad, M., Richi, T., Arafat, M., **Jiko, P. A.**, Haque, R., Hossain, H., Hossain, S., Ashrafi, S., Taher, A., Alam, S. (2025). Amelioration of hepatic injury through oxidative stress management employing methanolic extract of Crepe- ginger (*Cheilocostus speciosus* (J. Koenig) C. Specht) flower. *Journal of Chemistry & Biodiversity*. DOI: 10.1002/cbdv.202500261.
12. Mohammad, M., Richi, T., Hossain, R., Arafat, M., **Jiko, P.A.**, Emon, N.U., Rabbi, S.A.H., Khastagir, T., Hossain, M.H., Alam, S. (2024). Hepatoprotective experiment of Taro vegetable (*Colocasia esculenta* (L.) Schott) flower employing animal models by mitigating oxidative stress. *Pre-Print*. DOI: 10.21203/rs.3.rs-5400811/v1.
13. **Jiko, P.A.**, Mohammad, M., Richi, F. T., Islam, A., Alam, S., Abdullah, M., Taher, Shao, Chuxiao., Wang, Shuanghu., Geng, Peiwu., Mamun, A. A (2024). Anti-Inflammatory, Analgesic and Anti-Oxidant Effects of *Shirakiopsis Indica* (Willd). Fruit Extract: A Mangrove Species in the Field of Inflammation Research. *Journal of Inflammation Research*. DOI: 10.2147/JIR.S470835

MANUSCRIPT SUBMISSION (Under Peer Review)

No	Title	Journal	ID
1.	Food Preservatives and Red Blood Cell Toxicity: Molecular Mechanisms Linking Oxidative Stress, Membrane Damage, and Hemolysis	International Journal of Food Science	2019214
2.	Comparative Evaluation of Extraction Yield and Antimicrobial Activity of Six-Solvent Extracts from Different Color-Based Portions of <i>Colocasia esculenta</i> (L.) Schott Inflorescences	Journal of Tropical Medicine	3813229
3.	Comprehensive Evaluation of Methanol Extract of <i>Swintonia floribunda</i> Leaves on CCl ₄ -Induced Hepatic Injury, Renal Function Alterations, and Antidiabetic Activity in Swiss Albino Mice	Journal of Chemistry	5062288
4.	Solvent-Dependent Antimicrobial Activity of <i>Abroma augusta</i> and <i>Lasia spinosa</i> Extracts Against Human Pathogenic Bacteria: A Comparative In Vitro Assessment.	International Journal of Microbiology	5431425
5.	Anti-hyperuricemic and antioxidant potential of <i>Rhynchostylis retusa</i> leaf and root methanolic extracts in Potassium oxonate-induced hyperuricemia and renal injury in mouse model	BioMed Research International	6177490
6.	Pharmacological and Computational Assessment of Dragon Fruit Extracts for Antihyperlipidemic and Reno-protective Activities Using High Fat Diet and Nephrotoxicity Mice Models	Food Science & Nutrition	7210843
7.	Evaluation of Diuretic Potential and Urinary Electrolyte Excretion of <i>Artocarpus chaplasha</i> Seed methanolic Extract in Swiss albino mice	Animal Models and Experimental Medicine	1870411
8.	Pharmacological Profiling of <i>Firmiana colorata</i> (Roxb.) R. Br. Leaves: Linking Traditional Use with Experimental Validation of Hepatoprotective, Anti-Inflammatory, Antidiarrheal, and Anthelmintic Activities via ADMET Study and Molecular Docking	Biochemical and Biophysical Research Communications	BBRC-26-5958

CONFERENCE PRESENTATION (POSTER & ORAL)

- **Evaluation of the neuropharmacological and acute toxicological effects of *Colocasia esculenta* flower in experimental mice model**
Poster Presentation in BCSIR Congress-2023; 08-10 March 2024
- **The evaluation of antioxidant and improvement of CCl₄ induced hepatic injury of *Jatropha gossypifolia* Fruits extract in Swiss albino mice.**
Oral Presentation in BCSIR Congress-2023; 08-10 March 2024
- **A case study of the dangerous outbreak of Dengue in Chattogram: 2019**
Research-based Poster: Presented at 4th Innovations in Plant and Food Sciences (IPFS) and International Conference on Biotechnology in Health and Agriculture (ICBHA) at Nabab Nawab Ali Chowdhury Senate Bhaban of University of Dhaka. November 11-13; 2019
- **Present status and future management of Covid19 in Bangladesh & it's impact on youth mental health: Review and an online cross-sectional survey**
Online based Poster: Organizing by International Poster Presentation Competition (IPPC 2020) on 27th September 2020.
- **Human Papillomavirus and Cervical Cancer: A Systematic Review 2020**
Review based Poster: Presented at University of Science and Technology Chittagong (USTC).

TRAINING AND WORKSHOP

1. Training Courses on **Scientific Manuscript writing for Publication** at International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)
 - 8, 9 & 12 October 2025.
2. Training on **Basic Biotechnology** at the National Institute of Biotechnology (NIB).
 - January 04 to 09; 2020. (*Ref No. ST-001072*)
3. The 57th Understanding Training Courses on **ISO 15189:2022** (Medical Laboratories-Requirements for quality and competence) at the Bangladesh Accreditation Board (BAB); Ministry of Industry.
 - 11-13 March 2025 (*Ref. No. 25038905.121*)
4. Recognition for participating in **4th Innovations in Plant and Food Sciences (IPFS) and International Conference on Biotechnology in Health and Agriculture (ICBHA)** at Nabab Nawab Ali Chowdhury Senate Bhaban of the University of Dhaka.
 - November 11-13; 2019
5. Day-long Workshop on **CRISPR/Cas9 Genome Editing** Organized by the Global Network of Bangladeshi Biotechnologists (GNOBB)
 - 10th November, 2019.
6. Participated and Attended a Hands-on Training Course entitled “**Clinical Molecular Techniques in Disease Diagnosis**” at CVASU.
 - March 23-24; 2018
7. Participating 2nd National Workshop on **Bioinformatics and Computational Biology** held in Chittagong, Bangladesh.
 - December 18-21; 2016.
8. Field Training on **Clinical Biochemistry, Microbiology, Pathology & Blood Bank** at Chittagong Medical College and Hospital (CMCH).
 - July 05 to November 23, 2014. Reg. No. 23866 (*Ref. No. CMC/PG/2015/1307*)

PEER REVIEWER

- **Journal of annals and Medicine (Wolters Kluwer Health)** *September 2025 – Present*
- **Journal eFood (Wiley)** *May 2026 – present*
- **Chemistry and Diversity (Wiley)** *September 2026 – present*

PROJECT COORDINATOR

□ 2025 – Present • Research Assistant

Chittagong University of Engineering and Technology (CUET)

Under the supervision of **Prof. Dr. Ranajit Kumar Sutradhar**, Department of Chemistry

- Providing research and technical support for **four M.S. and three Ph.D. research projects**, contributing to project planning, experimental execution, data interpretation, and scientific reporting.
- Coordinating multidisciplinary research activities to ensure the timely achievement of project milestones and research objectives.
- Assisting in the preparation of research proposals, manuscripts, presentations, and other scholarly outputs.

□ 2024 – Present • Lead Coordinator and Co-PI

Bangladesh Bioscience Research Group (BBRG)

University of Science and Technology Chittagong (USTC); Faculty of Life Science; Department of Biochemistry and Biotechnology

- Supervised and mentored over **32+ undergraduate** and **17+ Post graduate students** in research methodology, experimental design, and scientific writing.
- Coordinated multidisciplinary research activities with **10+** national and international collaborators, including monthly project meetings, progress monitoring, and research seminars.
- Achieved 2 peer-reviewed publications and submitted 4 manuscripts over the past two years, while currently leading 7 multidisciplinary research projects at various stages of completion.
- Provide strategic guidance in research project management, fostering collaborative research environments and promoting high-quality scientific outputs.

□ 2022 – Present • Independent Collaborative Projects: On-going

- i. The knowledge, Attitude and Practice of Community People on Dengue Fever in Chattogram: A Cross-sectional Study.
- ii. Trends in hematological and biochemical profile of different clinical forms in seropositive dengue fever patients during the 2025 outbreak in south-eastern part of Bangladesh.
- iii. Immune Response and Molecular Insights of Dengue Fever: A Systematic Review
- iv. Epidemiological and Hematological Characterization of Inherited Hemoglobin Disorders in Chattogram, Bangladesh: Iron Status and Gender-Specific Variations.
- v. Antidiabetic Activity of Methanolic Fruit Extract of *Ficus hispida* in Alloxan-Induced Diabetic Mice.

FUNDING

□ Current Funding

1. Worked as **Lead Coordinator** in a UGC and CVASU funded project entitled '*Molecular Epidemiology and Genetic Diversity of the Circulating Dengue Serotypes in Chattogram: 2024 Outbreak*'. Under the supervision of Prof. Dr. Sharmin Chowdhury, Department of Pathology and Parasitology at CVASU.

□ Project Completed

2. Worked as **Co-PI** in University Grant (URC Call-05) at University of Science and Technology funded project titled as '*Triglyceride glucose (TyG) index and HbA1C Level as a biomarker for predicting prediabetes and cardiovascular risk in healthy young adults: A preliminary study.*'

3. Worked as **Project Coordinator** in partially funded by Chattogram Veterinary and Animal Sciences University and UGC of Bangladesh grant no.: 2023-24, 19 (to T.M.R.) projects titled as '*Integron-Mediated Convergence of Carbapenemase and Disinfectant Resistance in 2 Acinetobacter spp. from Critical Care Units.*'

LANGUAGES

Bengali (Native)

English (Professional Proficiency)

REFEREES

Dr. Nurul Absar; PhD <i>Faculty Dean and Professor</i> Faculty of Life Science University of Science and Technology Chittagong (USTC) +8801717 796 123 nurul_ustc@yahoo.com	Dr. Mohammad Shahangir Biswas; PhD <i>Head and Associate Professor</i> Department of Biochemistry and Biotechnology University of Science and Technology Chittagong (USTC) +8801323 381 081 shahangirbiswas@ustc.ac.bd
Dr. Ranajit Kumar Sutradhar; PhD <i>Professor</i> Department of Chemistry Chittagong University of Engineering & Technology (CUET) +8801812 149 299 rksutradhar2002@yahoo.com	Dr. Sharmin Chowdhury; PhD <i>Professor</i> Department of Pathology and Parasitology Faculty of Veterinary Medicine; Chattogram Veterinary and Animal Sciences University (CVASU) +8801554 331 355 sharminchowdhury@cvasu.ac.bd
Dr. Nasreen Chowdhury; FCPS; MD <i>Professor</i> Department of Biochemistry Bangladesh Medical University (Ex: Bangabandhu Sheikh Moji Rahman Medical University-BSMMU) +8801711 331 160 drnchowdhury71@gmail.com	Dr. Sayeed Mahmud; MPH; PhD <i>Professor</i> Department of Community Medicine Institute of Applied Health Science -USTC Course Coordinator, School of Science and Technology (BOU/CMCH) +8801819 296 301 sayeed12cmc@yahoo.com